

SB Care: Online Health Care Management and Appointment Portal for San Bartolome Health Center in San Bartolome, Santo Tomas, Pampanga

Introduction / Background

Barangay health centers provide basic healthcare services to communities and serve as the first point of contact for many patients seeking medical assistance. However, many health centers still rely on manual and paper-based systems for managing patient records, scheduling appointments, and documenting health services. These traditional methods often lead to long waiting times, slow service delivery, and difficulties in retrieving important patient information.

To address these challenges, this study proposes SB Care: Online Health Care Management and Appointment Portal for San Bartolome Health Center in San Bartolome, Santo Tomas, Pampanga. The system aims to provide patients with easier access to their medical records, medication lists, and appointment schedules through an online platform. It also includes an AI Consultation Assistant that provides basic health advice and recommends clinic visits when necessary.

Previous studies have shown that digital health information systems significantly improve healthcare services by making records more accurate, accessible, and organized. These systems also help healthcare workers generate reports faster and manage patient information more efficiently. However, some healthcare facilities still experience challenges such as outdated technology, limited training, and slow internet connectivity, which can affect the implementation of digital systems. Despite these challenges, the adoption of online healthcare management systems has proven to be beneficial in improving healthcare services in many communities.

Problem Statement

Many barangay health centers still rely on manual processes for managing patient records and healthcare services. This situation leads to several operational problems that affect both healthcare workers and patients.

One major issue is the long waiting time experienced by patients due to manual registration and record retrieval. Health workers often spend significant time searching for patient records in paper files, which slows down the service process. In some cases, paper records may become misplaced, damaged, or incorrectly documented, which can lead to errors in patient information.

Another problem is the lack of an organized appointment system, which can result in overcrowding and inefficient patient flow within the health center. Additionally, patients have limited access to their own health information, making

it difficult for them to monitor their medical history and medications. These challenges highlight the need for a more efficient and organized healthcare management system that can improve accessibility, reduce waiting time, and support healthcare workers in delivering better services.

Proposed Solution

To address the identified problems, this study proposes the development of SB Care: Online Health Care Management and Appointment Portal. The system is designed to improve healthcare service delivery by digitizing patient records and providing an online platform for managing appointments and healthcare information.

The proposed system includes a patient portal that allows users to access their medical records, view prescribed medications, and schedule appointments with the health center. By providing online access to these services, patients can better manage their healthcare and reduce the need for manual processes.

Another key feature of the system is the AI Consultation Assistant, which provides basic health advice for minor health concerns. The AI assistant can answer simple health-related questions and recommend whether the patient should visit the health center for further medical attention.

In addition, the system will include a Patient ID Pass system, which enables healthcare workers to quickly retrieve patient records by scanning the patient's identification pass. The portal will also display active health center staff, allowing patients to see which doctors or health workers are currently available in the facility. Through these features, the proposed system aims to improve efficiency, reduce manual workload, and provide faster and more organized healthcare services to the community.

Target Users

The primary users of the SB Care system include patients, doctors, health workers, and the local community. Patients will benefit from easier access to their medical records, medication lists, and appointment schedules through the online portal. The system also allows them to receive basic health advice from the AI Consultation Assistant for minor health concerns.

Doctors will benefit from the system by having faster access to patient information through the Patient ID Pass system, allowing them to focus more on providing medical care instead of handling manual paperwork. Health workers will also experience improved coordination and organization of tasks, particularly through the feature that shows active staff members in the health center.

Overall, the community will benefit from faster, more organized, and more accessible healthcare services provided by the health center.

Scope and Limitations

This study focuses on the development of SB Care: Online Health Care Management and Appointment Portal for San Bartolome Health Center in San Bartolome, Santo Tomas, Pampanga. The system is designed to help patients access their medical records, track their medications, and manage their appointments more conveniently through an online platform.

The system will also include an AI Consultation Assistant that provides advice for minor health concerns, a Patient ID Pass system that allows doctors to quickly retrieve patient records, and a portal feature that shows which doctors or staff members are currently active in the health center. The system will be tested within the health center to evaluate its functionality, usability, and effectiveness in improving healthcare services.

However, the study has several limitations. The system will only cover the services and operations of San Bartolome Health Center and will not include advanced medical diagnostic tools or emergency response services. It will also not be connected to external hospitals or laboratories. The AI Consultation Assistant is intended only for basic health guidance and cannot replace professional medical diagnosis. Additionally, the accessibility of the system may depend on the availability of internet connection and digital devices among patients.

Technology to be Used

PHP – to handle the server-side logic of the system.

HTML, CSS, and JavaScript – to design and build the website interface.

MySQL – to store and manage the system database.

phpMyAdmin – to manage and administer the MySQL database.

Visual Studio Code – as the main editor for writing and debugging code.

XAMPP – to run Apache and MySQL for local development and testing.